

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

A Popular P2P Network: BitTorrent

Definition

BitTorrent is a Peer-to-Peer (P2P) protocol designed by **Bram Cohen** for efficient sharing of large files among multiple peers.

Key Idea

- Entire file is divided into small pieces (chunks).
- Peers simultaneously
 - Download chunks from other peers.
 - Upload chunks to other peers.
- Sharing is collaborative rather than downloading from a single source.

Exam Point

BitTorrent follows a collaborative “download + upload” mechanism instead of a traditional client-server download.

Important Terminology

Torrent

A collaborative process in which peers exchange pieces of a content file.

- **Swarm**

- Collection of all peers participating in a torrent.

- **Seed**

- Peer possessing the complete file.

- **Leech**

- Peer having only part of the file and downloading the remaining pieces.

Key Point

A swarm consists of both **Seeds** and **Leeches**.

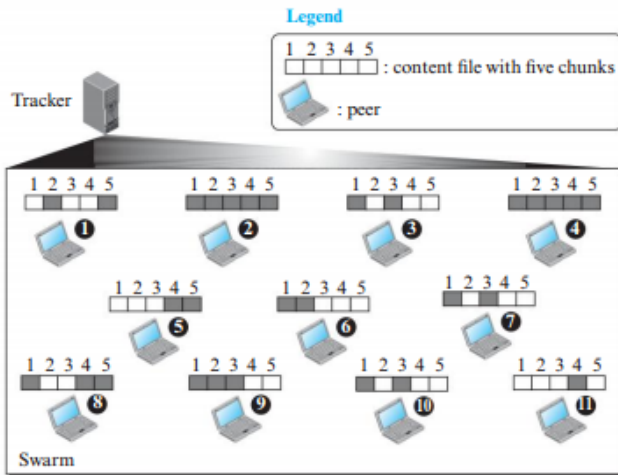
Tracker

A tracker is a central node that keeps track of peers participating in a torrent.

Working

- 1 Large content file is divided into chunks.
- 2 Seeds contain the complete file.
- 3 Leeches contain only some chunks.
- 4 Peers continuously upload and download chunks.
- 5 Peers may join or leave the torrent at any time.

Figure 2.57 *Example of a torrent*



Note: Peers 2 and 4 are seeds; others are leeches.

Joining a Torrent

How a New Peer Joins a Torrent

- 1 User requests the required content file.
- 2 BitTorrent server sends a **torrent file (metafile)**.
- 3 Torrent file contains
 - Information about file chunks
 - Address of the tracker
- 4 New peer contacts the tracker.
- 5 Tracker returns a list of neighbouring peers.
- 6 The peer joins the swarm and starts downloading and uploading chunks.
- 7 After obtaining all chunks, the peer becomes a **Seed**.

Key Point

A peer may remain in the swarm after downloading to help other peers.

BitTorrent Fairness Policy

BitTorrent follows several policies to ensure:

- Fair sharing among peers
- Efficient exchange of pieces
- Prevention of peer overload
- Better download performance

Neighbour Management

Each peer limits the number of concurrent neighbours.

- Typical value: **4 neighbours**
- Each neighbour is classified as:
 - Unchoked / Choked
 - Interested / Uninterested

Exam Point

Limiting neighbour connections prevents overloading and improves fairness.

Choking and Unchoking

Neighbour Categories

- **Unchoked Peers**
 - Currently connected
 - Uploading and downloading data
- **Choked Peers**
 - Not currently connected
 - May be selected later

Every 10 Seconds

- Peer tests one interested choked neighbour.
- If the new neighbour offers a better transfer rate, it replaces the slowest unchoked peer.

Key Idea

Peers continuously communicate with neighbours offering the highest data rates.

Optimistic Unchoking

- Every **30 seconds**
- One interested choked peer is randomly promoted.
- Promotion is independent of upload speed.
- Helps newly joined peers obtain their first pieces.

Rarest-First Strategy

- Download the least available chunks first.
- Rare chunks spread quickly through the swarm.
- Improves overall availability of file pieces.

Trackerless BitTorrent

Need for Trackerless BitTorrent

If the central tracker fails,

- New peers cannot join.
- Swarm information cannot be updated.

Solution

- Use a **Distributed Hash Table (DHT)**.
- Tracking responsibility is distributed among peers.
- Kademlia DHT locates the node responsible for a torrent.
- Responsible node returns the list of peers.
- BitTorrent then performs normal chunk sharing.

Exam Point

Trackerless BitTorrent removes the single point of failure by replacing the central tracker with a Distributed Hash Table (DHT).

BitTorrent – Key Takeaways

Summary

- BitTorrent Protocol
- Torrent
- Swarm
- Seed
- Leech
- Tracker
- Torrent File (Metafile)
- Choking & Unchoking
- Optimistic Unchoking
- Rarest-First Strategy
- Trackerless BitTorrent
- Kademlia DHT

Frequently Asked KTU Exam Topics

- Explain the working of BitTorrent.
- Define Torrent, Swarm, Seed and Leech.
- Explain Choking and Unchoking.
- What is Optimistic Unchoking?
- Explain the Rarest-First strategy.